

Operating System Projects

1/2025 edition

General Information

- Group of up to 6 students
- Submit separate reports/slides for each project assessment activity on LEB2.
- Reference to the sources of your knowledge, failure to do so result in deduction of your score.

Project #1

- Installation of a Linux system
- Aim on the alternative operating system experience
- You are going to—
 - Provide a study on popular Linux distributions
 - Explain which Linux distribution is your target and why you pick it
 - Provide a step-by-step explanation with screenshots of how to Install Linux systems on VMs
 - Explore the options on Linux installation
- What to submit —
 - A tutorial video on how to install your Linux of your choice
 - A slide accompany the presentation addresses the learning outcomes

Project #2

- Linux kernel module
- Aim on the operating system kernel module experience
- You are going to—
 - Explain the functionality of a kernel module
 - Choose what you want to do with the kernel module
 - Provide a step-by-step explanation with screenshots of how you can **create/modify/improve** the selected kernel module
 - Provide a step-by-step explanation with screenshots of how you can **compile** and **use** the selected kernel module
 - Provide evidences that your modifications have been applied
- What to submit —
 - A report on how to use Linux kernel modules

Project #3

- A case study of an Operating System Topics
- Create a report and a video presentation for one of the topic list
- The technical aspects aspect requirements are listed on the next page
- What to submit –
 - a report + a video presentation on the selected topic.

Topic list

Storage and Persistence

- Solid State Drive (SSD)
- GPT and UEFI
- Ventoy
- Clonezilla
- FAT32, exFAT, NTFS, and ext4 File Systems
- Network File Systems (NFS)

Virtualization and Infrastructure

- Proxmox VE
- VirtualBox

- VMware
- Windows Subsystem for Linux (WSL2)
- Docker and Containers

Advanced Operating Systems

- Real-Time Operating Systems
- Distributed Operating Systems
- Embedded Operating Systems
- Advances in Operating Systems

Technical Investigation Requirement

Students must include at least one of:

- Installation
- Configuration
- Experiment
- Demonstration
- Performance comparison
- Architecture analysis

Examples:

- Ventoy → Create and analyze a bootable USB
- Proxmox → Install and create VMs
- GPT/UEFI → Analyze disk layouts from real systems
- SSD → Compare SSD technologies and characteristics
- WSL2 → Compare WSL2 and native Linux